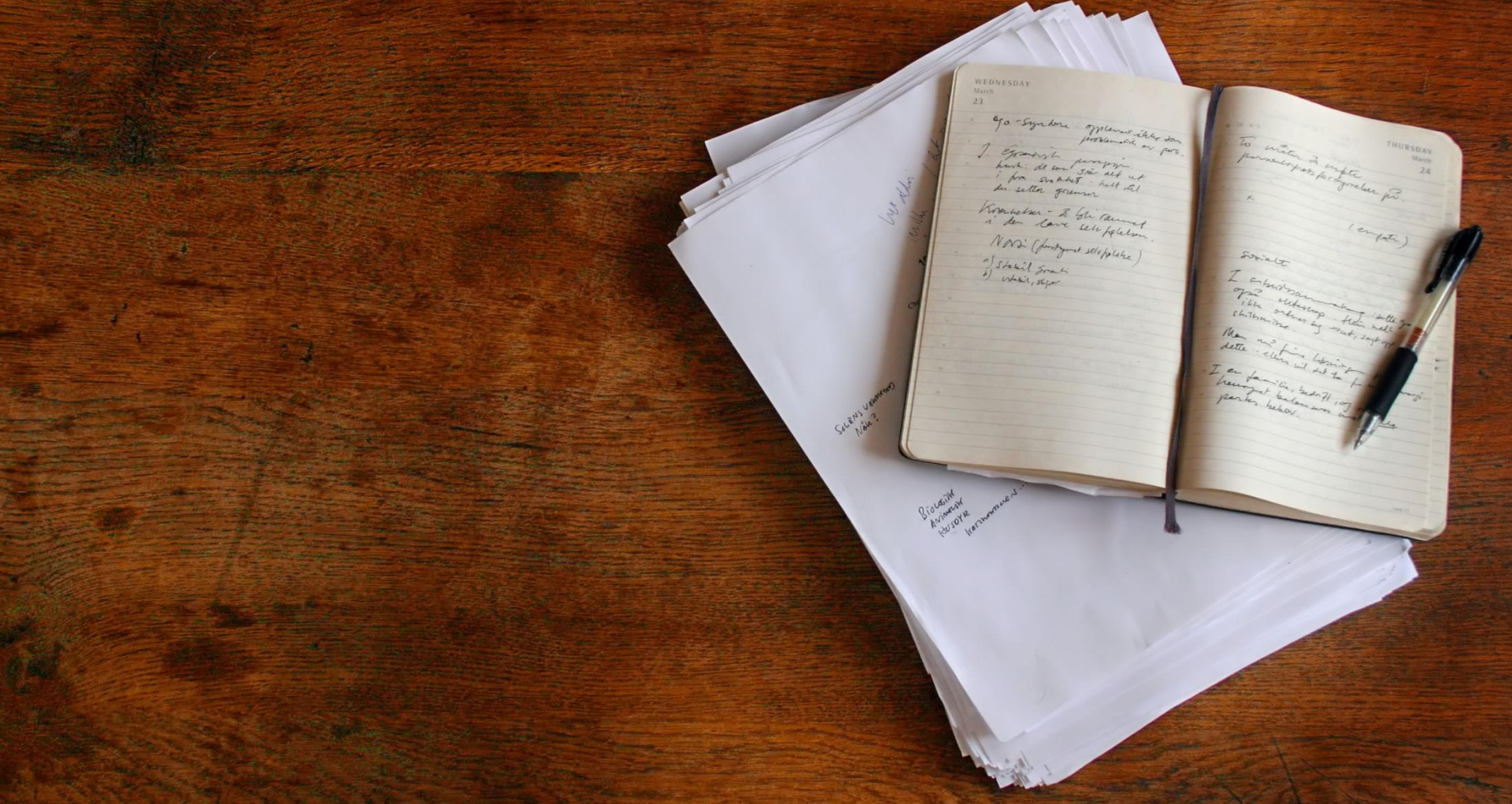




ACADEMIC WRITING (BETA) LECTURE, WEEK 3

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WHAT 'STORIES' DO ACADEMIC TEXTS TELL?



Solve a problem

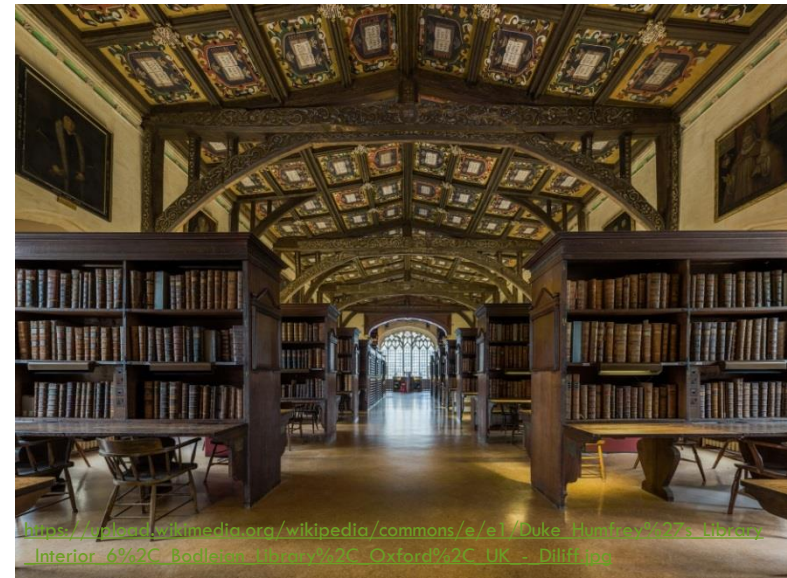
Present a new idea

Find a solution



Explain what research has been done so far

Report on an experiment





Index Terms—user study, security policy, risk compensation, two-factor authentication

I. INTRODUCTION

Passwords remain the weakest component of many important security systems, so there is a concerted push from

Abstract—Choosing the security architecture and policies for a system is a demanding task that must be informed by an understanding of user behavior. We investigate the hypothesis that adding visible security features to a system increases user confidence in the security of a system and thereby causes users to reduce how much effort they expend in other security areas. In our study, 96 volunteers each created a pair of accounts, one secured only by a password and one secured by both a password and a fingerprint reader. Our results strongly support our hypothesis—on average, when using the fingerprint reader, users created passwords that would take one-third as long to break, thereby potentially negating the security advantage that two-factor authentication could have offered.

Index Terms—user study, security policy, risk compensation, two-factor authentication

I. INTRODUCTION

Passwords remain the weakest component of many important security systems, so there is a concerted push from many directions to supplant or supplement less-fragile security measures. While this push has had some effects, particularly in environments that require authentication, it has failed to replace passwords—the vast majority of computer users still use passwords on a day-to-day basis. The security of passwords relies so heavily on user behavior, studies that empirically examine patterns of password creation and use remain important in the evaluation of security policies.

Most empirical studies of user password behavior provide baseline information about how users create, remember, or use passwords [1]–[3] or the kind of passwords users create [4]. Some offer practical results based on comparing user groups given different sets of advice [5] or by persuading users to use stronger passwords [6]. A larger number of user studies have been done on alternatives to passwords, such as graphical passwords [7] or biometric authentication [8], though most of these have studied usability or subjective preferences rather than the security-related behavior of the users. Our study fills a void in the examination of a multi-factor authentication scheme by reviewing how users behave in such a system compared to how they behave in similar systems that use only single-factor authentication.

When policymakers decide to add to or improve upon password authentication rather than replace it with a different authentication scheme, the default assumption is that the additional layer or layers of security are essentially independent of other layers; that adding a security mechanism to a system

will not decrease the overall security of that system. From a software perspective, this assumption is generally true and can usually be verified. However, the security of a system is inextricably linked to and dependent upon the behavior of the authorized users of that system. If adding visible security features increases the level of confidence those users have in the security of the system, they may decide to expend less effort in behaving securely. This is an outcome predicted by risk compensation theory [9] or risk homeostasis theory [10].

good security.

II. PREVIOUS WORK

As mentioned in the introduction, many studies have been done to establish the baseline behavior of users, including the

user assigned to their accounts (the ratio of money put in each account), we asked the users directly about their confidence in the security measures protecting the two accounts. Finally, we asked some demographic questions in order to assess potential differences between user groups. We also observed users to see how many wrote down their passwords, but these data are unpaired to user accounts, so we cannot draw conclusions from it.

In order to analyze the results of our study we measured the strength of user passwords using a hybrid metric that combines the number of guesses needed to guess a password. We used entropy measurements in this metric derived primarily from the performance of several password-cracking techniques on each password. As a measure of strength, we found a large variance in the strength of each participant's pair of passwords was. Across our entire study, the passwords connected to financial accounts were $\frac{1}{25000}$ the strength of password-only accounts. The decrease in strength was more marked for several subgroups within our study. We found strong correlations between a decrease in password strength and an expectation of the fingerprint reader providing

security.

III. PREVIOUS WORK

As mentioned in the introduction, many studies have been done to establish the baseline behavior of users, including the extremely large study presented by Zviran and Haga [11] of over half a million users. Moving from those baselines to study the effects of other policies on user behaviors, a number of empirical studies have determined factors that may improve the quality of user passwords, such as the study of 288 Cambridge Natural Science first-year students [5]. Other empirical studies have developed systems to improve the passwords users produce, such as the Persuasive Text Password system developed by and tested on 83 students of Carleton University in Ottawa [12].

However, the evaluation of policies designed to improve user passwords is significantly different from the purpose of our work, which aims to determine the unintentional and potentially negative consequences of policies and security architecture. Research has been done into unintended negative consequences of password policies [13], determining that

passwords [14]; reduces [15]. authentication, they found that users of our system, our vice in. data and posture is of a develops

suggestions of how to phrase arguments to persuade users to adopt better security behavior. Stanton et al. [18] likewise interviewed users for the purpose of discovering the range of user security behavior in a wide variety of contexts—not just in password choice and use. The laboratory-based user studies that the authors are familiar with mostly attempt to test new security systems, such as the evaluation of D6 [19].

experiment to determine the presence and strength of this effect in password choice.

III. METHODOLOGY

Our experiment consisted of a user study conducted in a laboratory setting that asked users to create two accounts each; we recorded the two passwords each user created. We describe the design of the user study itself in Section III-A. For analysis of the collected data, adequately measuring the strength of passwords collected in this study was of primary importance. We discuss what such measurement means in Section III-B.

The methods that we decided to use in assessing the strength of the collected passwords included various modes of John the Ripper, discussed in Section III-D, and two analytic measures. Our use of John the Ripper required password dictionaries described in Section III-C; these password dictionaries were also used to establish a baseline expectation about the distribution of password strengths and are therefore themselves relevant in our later analysis. The analytic measures we used to gauge password strength were class size estimation, discussed in Section III-E and entropy, discussed in Section III-F.

A. Experiment Procedure

In order to determine how our subjects chose passwords in different security environments, we wrote a desktop application that guided each volunteer through an account creation process, similar to creating an account on a website. Each volunteer was asked to create two accounts: a “standard” account secured only by a password and a “biometric” account secured by both a password and a fingerprint. The order that volunteers created accounts was randomized between subjects. After they had created both accounts, they were asked questions to assess how secure they felt the two accounts were, as well as questions regarding their security background.

1) *Recruitment*: After we received approval from our Institutional Review Board (IRB) to run a psychological study with human participants, we began recruiting students. Our volunteers were recruited primarily through fliers posted around our college campus offering \$5 to participate in an experiment to test hackers. Some volunteers heard about our experiment through word of mouth, and toward the end of the experiment we set up a booth with a large poster outside the dining hall in the student union building.

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TODAY: QUOTING AND REFERRING

1. When do you refer to other articles?
2. How do you refer to other articles?



**WHEN DO YOU REFER?
(FOR WHAT REASON)**

I.

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I. TO EXPLAIN WHAT OTHER RESEARCHERS HAVE DONE

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II.

It has been argued that certain qualities present in the medium of videogames can provide valuable opportunities for learning [10, 36]. Furthermore, games are a unique medium because they present a new form of persuasive rhetoric [3]. In what ways can we use games to make moral demands of players encouraging them to reflect on ethical issues? Ultimately, what role can games play in help us become better people?

II. TO PROVIDE THE STARTING POINT FOR A DISCUSSION OR EXPERIMENT

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III.

The Internet of Things concerns more than just internet for computers; it concerns all devices that are connected to both the internet and each other [1].

Facebook is the largest database of social information in the world and gives the opportunity to share information with Facebook friends [2].

Show that you did not make this up.

III. TO GIVE DEFINITIONS AND FACTS

The Internet of Things concerns more than just internet for computers; it concerns all devices that are connected to both the internet and each other [1].

Facebook is the largest database of social information in the world and gives the opportunity to share information with Facebook friends. [2]

English is frequently used in TV commercials and is even the most used language in every segment of advertising (Pagani et al., 2015).

According to the Independent Regulator and competition authority for the UK communications industries, 93% of the adult people own a mobile phone, with which they send 117 text messages per month.



**WHEN DO YOU REFER?
(IN WHAT PART OF A PAPER)**

REFERENCES PER SECTION

I		
M		
R		
a		
D		

REFERENCES PER SECTION

I	Introduction + Previous research	
M	Methods	
R	Results	
a		
D	Discussion + Conclusion	

EXAMPLE PAPER: INTRODUCTION

I. INTRODUCTION

Crowd behaviour is an interesting social phenomenon that emerges from complex interactions of individuals. The analysis and study of crowds is a multi-disciplinary problem with a wide range of useful applications, including egress planning, event planning and structure design. Modelling and simulation is one very common way of understanding and predicting the behaviour of human crowds. Current literature describes various approaches and methodologies for modelling crowds, such as flow models [1], force models [2] and agent-based models [3]. The latter defines behaviours and rules at the level of each individual. This allows modellers to describe individual behavioural and psychological characteristics. These models are theoretically able to describe a variety of different factors relating to human behaviour, such as gender, age, cultural variation, emotional characteristics, etc.

EXAMPLE PAPER: PROPOSAL OF MODEL

The emotion engine proposed here is based on appraisal theory [6]. In particular, we make use of the component process model of emotion as described by Scherer [13].

In general the proposed emotion engine works as follows. Each agent is equipped with a perception system which perceives facts and events which are mapped to corresponding appraisal patterns $\mathbf{A} \in \mathbb{A}^m$, where $\mathbb{A}$ is the appraisal value domain and m the number of appraisal variables. The resulting m -dimensional appraisal space $\mathbb{A}^m$ is thus the space of all possible appraisal patterns. More formally, we use a perception-appraisal mapping function f_{pa} to map a value from the perception space $\mathbb{P}$ to the appraisal space $\mathbb{A}^m$:

$$f_{pa} : \mathbb{P} \rightarrow \mathbb{A}^m \quad (1)$$

EXAMPLE PAPER: EXPERIMENT (METHODS)

A total of 16 players were surveyed. The participants were not told about the difference between the missions (if any). After completing both missions, each participant was asked to complete the survey form. Participants were divided into three groups:

- Group 1: 5 participants playing the difficult mission first, then the easier mission.
- Group 2: 5 participants playing the easier mission first, then the more difficult mission.
- Group 3: 6 participants playing either the difficult or the easier mission twice.

Every mission was played for exactly 5 minutes regardless of whether any protesters managed to enter the property. As for the experimental setup, in each mission the participant was confronted with a crowd comprising of 126 protesters (either peaceable or bellicose) and 5 instigators. In addition to the survey we also recorded the time when the first trespassing occurred for each mission (i.e., when the first protester entered the restricted zone). For the difficult mission, no player managed to prevent protesters from trespassing. For the

EXAMPLE PAPER: EXPERIMENT (RESULTS)

These results are backed by the survey. A crowd that is made of bellicose agents is expected to be more difficult to control because of the increased anger level. Almost all participants, except for one, in Group 1 and Group 2 (i.e., 9 out of 10 participants) noticed a difference in difficulty and correctly identified the missions which were supposed to be easier. As for the control group (i.e., Group 3), 4 out of 6 participants (correctly) reported that they noticed no difference in difficulty. These results indicate that participants were able to correctly tell whether or not there was a difference in difficulty (i.e., by performing a hypothesis test on the difference in the proportions of participants that *correctly* perceived whether there was any difference in difficulty, $H_0: p_1 = p_2$ cannot be rejected with 95% confidence where $p_1 = 9/10$ and $p_2 = 4/6$). In addition, 2 out of 6 participants in the control group reported that the second mission was easier to accomplish. We attribute this to the fact that players were more familiar with the game after playing the first mission. This has also been reported by one of those two players.

EXAMPLE PAPER: CONCLUSION

In this paper we have introduced our emotion engine based on modern appraisal theory. In contrast to closely related work, our engine has been designed for the application of crowd simulation in the context of serious games. We have introduced the concept of emotion triggers and explained how they can be used to realise emotional contagion and reflect a pacifying soldier action. In addition, we have explained how different character types can be simulated by using different susceptibilities and emotion half-lives. The effect of these different character types is reflected in the crowd behaviour. A bellicose crowd is more likely to push forward and trespass the area of the protected facility. Our experimental results have indicated a clear relationship between the difficulty of the game and the characteristics of the crowd.

REFERENCES PER SECTION

I	Introduction + Previous research	Many
M	Methods	Some (e.g. if method close to other work)
R	Results	Few (sometimes to compare results)
a		
D	Discussion + Conclusion	Some (relate results to previous research, advantages of this paper)

USING SOURCES IN THIS COURSE

Assignments 2-3:

- Make sure you include a bibliography and in-text references

Assignment 4:

- Introduction: an introduction to the topic, with reference to other research on the topic
- Middle part ('body'): discussion/comparison of 2-5 articles
- Conclusion: no new references
- Make sure you include a bibliography and in-text references



HOW DO YOU REFER?

REFERENCING STYLES: TWO MAIN ASPECTS

1. In-text references
2. Bibliography

3 LEVELS OF REFERRING

Very common in introductions

1. Referring (only the main point)

There has been a debate on which aspect of language is most important in language acquisition (Smith 2013, Jones 2012).

Most common form

2. Paraphrasing (at different levels of detail)

Smith (2013) concludes that syntax is the most important aspect in language acquisition. Jones (2012), on the other hand, argues that it is semantics that plays the determining role.

Not often used in your disciplines

3. Quoting (in detail)

Smith (2013) says that “syntax is crucial in language acquisition” (2013: 45). Jones (2013) defines language as “a collection of sentences” (2013: 13)

WHAT INFORMATION TO INCLUDE?

What information do you need to find a book?

What information do you need to find an article?

AT ALL TIMES, REMEMBER THE PURPOSE OF PUTTING IN THE REFERENCES



Google

You shouldn't need Sherlock to find the source. And don't expect your reader to rely on Google only.

<http://i.ebayimg.com/images/g/-NwAAMXQ76RR5XqC/s-l300.jpg>
http://img.thesun.co.uk/aidemitlum/archive/01437/SNF1428A--532k_1437565a.jpg

IN THIS COURSE — REFERENCING STYLES

1. IEEE
2. Harvard

STYLE 1: IN-TEXT REFERENCES

A survey by the United Nations for example has reported that “robots are set to become increasingly familiar companions in the home by 2007.” By 2007 it is predicted that there will be almost 2.5 million entertainment and “leisure” robots in homes which compares to 137,000 currently [1].

However, increasingly robots are meant to engage in social-human interaction including e.g. [3], [4], [5], [6].

The studies by Khan [17] and Scopelliti, Giuliani, D’Amico and Fornara [18] are among the first to have used a psychological design framework using questionnaires to explore adults attitudes towards the design of a domestic robot.

STYLE 1: LIST OF REFERENCES

- [1] B. News, "Robots set to get homely by 2007," in <http://news.bbc.co.uk/go/pr/fr/-/2/hi/technology/3764142.stm>, 2004.
- [2] D. Wilkes, R. Alford, R. Pack, R. Rogers, R. Peters, and K. Kawamura, "Toward socially intelligent service robots," *Applied Artificial Intelligence Journal*, vol. 12, pp. 729-766, 1997.
- [3] Sony, "Sony Entertainment Robot Europe <http://www.aiboeurope.com/>," 2004.

STYLE 2: IN-TEXT REFERENCES

The importance of teams as fundamental units of work in organizations (Hackman 1987, Sproull and Kiesler 1991) has been well-documented (Cohen and Bailey 1997, Harrison et al. 2003, McGrath 1991).

As Simon's (1955) concept of "bounded rationality" implies ...

Brandon and Hollingshead (2004) have suggested that ...

STYLE 2: LIST OF REFERENCES

Aiken, L., S. West. 1991. *Multiple Regression: Testing and Interpreting Interactions*. Sage, Newbury Park, CA.

Alavi, M., D. E. Leidner. 2001. Knowledge management and knowledge management systems: Conceptual foundations and research issues. *MIS Quart.* 25(1) 107–136.

Alker, H., M. Russet. 1964. On measuring inequality. *Behavioral Sci.* 9(3) 207–218.

From: Espinosa, J.A., S.A. Slaughter, R.E. Kraut & J.D. Herbsleb (2007). Familiarity, Complexity, and Team Performance in Geographically Distributed Software Development.





QUIZTIME

Take a pen and paper, or your laptop, and write down the answers to the following questions:

1. When do you refer to other articles? Give at least two reasons.
2. What information needs to be included in the bibliography? Give at least 4 items.

COMING UP NEXT...

1. This week's seminar:

- Practising with paraphrasing, summarizing and quoting
- Practising with referencing style
- Practising with avoiding plagiarism

2. Next week:

- Friday 25 April: Hand in Assignment 2
- Lecture: Paragraph structure
- Seminars: Paragraph structure, linking language and clause combining